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| **Fiscal Year:** | FY 2012 | **Task Last Updated:** | FY 02/14/2012 |
| **PI Name:** | Ziegler, James Ph.D. | | |
| **Project Title:** | Lunar EVA Dosimetry: Microdosimeter-Dosimeter Instrument | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI--Radiation Effects Team | | |
| **Joint Agency Name:** | | **TechPort:** | Yes |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Chester | **State:** | MD |
| **Zip Code:** | 21619 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2007 Crew Health NNJ07ZSA002N |
| **Start Date:** | 10/01/2011 | **End Date:** | 09/30/2013 |
| **No. of Post Docs:** | 3 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 5 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 4 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 9/30/2013 per NSBRI (Ed., 3/12/13)<br>NOTE: Original task began 1/1/2009 with PI V. Pisacane. Period of performance changed with new PI, James Ziegler, to 10/1/11-3/31/13 (Ed., 2/29/2012) | | |
| **Key Personnel Changes/Previous PI:** | Former PI Vincent Pisacane retired as of 9/30/2011; James Ziegler is new PI effective 10/1/2011 and project continues through 3/31/2013, per NSBRI. See Pisacane for FY2011 and earlier reports (Ed., 2/29/2012) | | |
| **COI Name (Institution):** | Cucinotta, Francis ( NASA Johnson Space Center )<br>Dicello, John ( United States Naval Academy )<br>Rozenfeld, Anatoly ( University of Wollongong )<br>Nelson, Martin ( United States Naval Academy )<br>Zaider, Marco ( Memorial Sloan-Kettering Cancer Institute )<br>Dolecek, Quentin ( United States Naval Academy ) | | |
| **Grant/Contract No.:** | NCC 9-58-RE01601 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | 1. PROJECT AIMS/OBJECTIVES. The objective is to advance the state-of-the-art of semiconductor micro-dosimeters (SMD) to design, develop, and test a flight-qualifiable engineering model by Dec 2012. This 4 year project started in 1 Jan. 2009 with this report concluding the 3rd year. Aims include: 1.1 - Develop a bench-top system to advance the state-of-the-art of Silicon micro-dosimeters (SMDs) to incorporate proven advancements into the flight engineering model; 1.2 - Develop a flight engineering model suitable as a personal SMD; 1.3 - Develop improved SMD sensors; 1.4 - Utilize computer modeling to support instrument development; 1.5 - Explore opportunities for transition to a flight program.<br>2. KEY FINDINGS. 2.1 System calibration without Radioactive Materials - A reliable new technology was developed, tested and a NASA Patent was filed for this technology. 2.2 Space Missions - The MIcro-Dosimeter Nucleon (MIDN – the acronym used in this project for all our micro-dosimeter systems) was launched on the MIDSTAR-I spacecraft in March 2007. It is the only semiconductor micro-dosimeter ever flown in space. Because of observed problems, a 2nd instrument, MIDN-III, is under development based. Improvements to the instrument include: • Increased use of rad-hard digital electronics in place of analog; • Power changed to rechargeable dual batteries to reduce noise; • Increased number of remote commands; • Improved SOI sensors. 2.3 Radiation Testing of MIDN micro-dosimeters – Extensive testing of new MIDN sensors was carried out with low energy protons, alphas and neutrons, and at NASA Space Radiation Laboratory-Brookhaven Nat'l Lab (NASA/BNL) facility using MeV/GeV protons and heavy ions. Concurrently, extensive modeling was done with GEANT4. The sensors showed a detection accuracy of better than 20% for most beams, with the GeV heavy-ions showing modest sensor saturation. No measureable degradation of the MIDN sensors. 2.4 Microdosimetry with Coincident Particle Identification - An experimental system for identifying particle types, energies, and mass-to-charge ratios of beams of energetic particles responsible for events of energy deposition. 2.5 New Radiation Sensor Designs – Investigation of the response new p-SOI radiation sensors was completed. They had good performance for 2MeV protons and 3MeV alphas. The noise is 4x better than MIDN-II, with noise levels of 0.07 keV/micron (~0.04 keV/micron-tissue). The arrays showed >99% functionality, and absence of cross talk between neighbor SVs qualifying the MIDN-III for higher level production. Other tests used MeV He and C-12 ions. Excellent performance of sensors was demonstrated. 2.6 Testing Ultra-Small Micro-dosimeters – Evaluation sensors with 1 µm dimensions have been obtained and experiments begun on the critical parameters: dose accuracy, background noise, saturation levels, and system robustness in a radiation environment.<br><br>3. IMPACTS. Silicon micro-dosimeters (SMDs) have been shown to have noise levels better than that obtained with TEPCs in space. SMD spectra for space protons will be able to obtain very low-lineal energy detection, a major goal of this research project. 2011 measurements of SMD spectra with intense high-energy neutrons (~14 MeV) show that SMD can operate in high-dose radiation fields for long time periods without failures. This establishes the radiation resistance of our SDMs, a major goal. 2011 measurements with SDM systems at the NSRL facility at BNL establish the practicality of using our new capability of identifying particle species: i.e. energy and charge-to-mass ratio responsible for specific individual events. Such measurements provide more stringent data for establishing quality factors and the accuracy of the transport codes, a major aim. Recent measurements with SMD in a plastic phantom on a HIMAC beamline (in Japan) demonstrated the success of MIDN-III sensors for deep space missions by showing equivalent performance to TEPC instruments. Development of an end-to-end system test and calibration of an astronaut's personal SMD without the need for an ionizing radiation source is an important achievement. The final testing of MIDN-II sensors, and the design and highly-successful early testing of the MIDN-III sensors (probably final flight qualifiable personal SMDs) are important accomplishments.<br><br>4.0 RESEARCH PLAN for 2012. 4-1. Obtain MIDN-III microdosimeter sensors. Obtain commercial 1um SOI detectors. 4-2. Design and build evaluation/test electronic sensor analysis systems. 4-3. Complete electronic testing on all 9 MIDN sensor technologies, and 5 commercial sensors. Evaluate and select the top 2 models for inclusion in the final prototype system. 4-4. Test final SMD selections for accuracy with particle beams: protons, alphas and neutrons. Compare with computer modeling to determine sensor accuracy. Determine any sensor non-linearity or signal saturation. Determine sensor reliability under extended radiation. 4-5. From radiation testing, search for sensor blooming (a single particle causing events in several sensors), signal saturation, and non-linear effects. 4-6. Establish final space-suitable dosimetry system goals and required displays. 4-7. Establish communication links between astronaut suit system and base-station monitor. 4-8. Build two complete flight engineering prototypes: (1) Astronaut suit system with helmet warning, and (2) Station monitoring system. 4-9. Test systems for sensitivity to external vibration, mechanical ruggedness and sensitivity to external RF radiation.. 4-10. Run ground tests for power consumption and establish suitable power packs. 4-11. Test prototypes with MeV ions. 4-12. Delivery of final flight engineering model with calibration spectra and microdosimetric spectra to NASA/NSBRI. 4-13. Flight Opportunity – Continue to work with the DoD Space Test Program Office to obtain a future flight opportunity. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | To determine the risk from currently used radiation dosimeters requires knowledge of the species, energies, and frequencies of the radiation types or the frequency distributions as a function of linear energy transfer. The more frequently used passive dosimeters in addition are processed after the exposure and are not real-time instruments so the risk is inferred only after exposure. Micro-dosimeters are unique in that they can be used to directly determine the regulatory risk from radiation in real time when neither the species nor energies of the radiation are known. Thus it is a superior instrument for use in situations when the radiation environment is unknown and perhaps time varying. With sufficient investment in VLSI technology the semiconductor micro-dosimeter can be integrated into a cell-phone sized instrument. Since microdosimetry provides the regulatory risks from radiation exposure in real time, it can be beneficially used by first responders in emergency situations when there is uncertainty in the radiation risk. The micro-dosimeter can be used in nuclear power plants and other facilities with radioactive materials to provide risk due to exposure. It can also be used to detected contraband radioactive material; because of its compact size and potentially relatively low cost, it can be used in situations where large numbers of sensitive detectors are needed. Development of SOI micro-dosimeters has a potentially significant impact on applications to monitor the dose equivalent during proton therapy to reduce the possibility of secondary cancers generated in normal tissue by the radiation. Development of a calibration technique that does not use an ionizing radiation source will reduce the exposure of handlers of the micro-dosimeter. It will also eliminate the cost of satisfying the regulations on certification of users and on the handling, shipping, and facilities. |

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| **Task Progress:** | NOTE: Former PI Vincent Pisacane retired as of 9/30/2011; James Ziegler is new PI effective 10/1/2011 and project continues through 3/31/2013, per NSBRI. See Pisacane for FY2011 and earlier reports (Ed., 2/29/2012)<br>The overall objective of this research project is to design, develop, and test an engineering model breadboard/prototype semiconductor micro-dosimeter by December 2012 suitable for use in the new NASA spacesuit and robotic operation on rovers, tool boxes, and spacecraft. The bench-top instrument continues to be used to investigate improvements to the state-of-the-art of semiconductor microdosimetry. This past year a focus has been on the development of an improved preamplifier. Radiation sources available at the USNA have been used to carry out the test protocols. The bench-top system has been expanded to obtain and analyze microdosimetric spectra for incident NSRL beams of both protons and heavy ions with identification of particle types in the beam, their energies, and their mass-to-charge ratios and those produced in intervening materials. We carried out tests of our bench-top system with a neutron beam generated in the Nucleonics Laboratory at the Naval Academy with favorable results. An improved version of the flight engineering model, MIDN-III, has been designed and is nearing completion. It has a reduced footprint and expanded remote command capability. It will be available for test by the end of the year. We processed data sets obtained using GeV ion beams at NSRL/BNL for our bench-top system, flight engineering model MIDN-III, and two Far West HAWK tissue equivalent proportional counters. Inter-comparisons of the observations agreed well and also agreed with Geant4 simulations. These spectra have been added to our past data sets to update our extensive library of microdosimetric spectra. We continued development our unique optical calibration system for a semiconductor micro-dosimeter that permits continual end-to-end system test and calibration while the instrument is operational deployed. This is an alternative to using a radiation source that is problematic in a personal dosimeter and eliminates handling and shipping restrictions and personnel and facility certifications required by international, federal, and local regulations. Our provisional patent application was followed by a full patent application. We have tested our 2nd generation microdosimeter sensors with our bench-top and flight engineering instruments and compared our results favorably with those obtained at the University of Wollongong. We received a sample of our third generation semiconductor micro-dosimeter sensors in November and will begin testing at the beginning of 2012. We completed the engineering design of our instrument to fit within a NanoRacks configuration for deployment on the International Space Station through the auspices of the DoD Space Test Program. The NanoRacks configuration is modeled after the design of a cubesat. Our configuration would be 10cmx10cmx15cm with the majority of the volume for a rechargeable battery power supply. Initial construction has been complete of the mechanical components. |
| **Bibliography Type:** | Description: (Last Updated: 02/25/2021) |
| **Articles in Other Journals or Periodicals** | Livingstone J, Prokopovich DA, Dzurak M Jamieson D, Perevertaylo VL,Kryukov A, Pisacane VL, Zaider M, Dicello J, Rosenfeld AB. "Comparison of Epitaxial and SOI based Large Area Microdosimeters." To be published in IEEE Transactions on Nuclear Science. December 2011. , Dec-2011 |
| **Articles in Peer-reviewed Journals** | Pisacane VL, Dolecek QE, Malak H, Cucinotta FA, Zaider M, Rosenfeld AB, Rusek A, Sivertz M, Dicello JF. "Microdosemeter instrument (MIDN) for assessing risk in space." Radiation Protection Dosimetry. 2011 Feb;143(2-4):398-401. Epub 2011 Jan 2. PubMed PMID: 21199825 , Feb-2011 |
| **Articles in Peer-reviewed Journals** | Pisacane VL, Dolecek QE, Rosenfeld AB, Malak H. "Proton and iron ion observations from a solid-state microdosimeter." Radiation Protection Dosimetry. 2012 Aug;151(1):117-28. Epub 2011 Dec 7. https://doi.org/10.1093/rpd/ncr452 ; PMID: 22155752 , Aug-2012 |
| **Articles in Peer-reviewed Journals** | Dicello JF, Pisacane VL, Dolecek QE, Rosenfeld AB, Rusek A, Sivertz M, Malak H. "Feasibility study of solid-state microdosimetry for energetic protons and heavy ions with coincident particle identification." Radiation Measurements, 2011 Dec;46(12):1539-42. http://dx.doi.org/10.1016/j.radmeas.2011.05.036 , Dec-2011 |